	Shantanu Edgaonkar	**22B2172**
	Mechanical Engineering	**B.Tech.**
	Indian Institute of Technology Bombay	**Gender: Male**
		DOB: 03/04/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	HSE	Maharashtra Technical Education Society Junior College	2022	
Matriculation	CBSE	City International School Kothrud	2020	

Scholastic Achievements

- Achieved an outstanding **99.68** percentile in **JEE Main**, surpassing nearly **1 million** students *(2022)*
- Secured All India Rank **601** in **JEE Advanced**, competing against **0.15+ million** candidates *(2022)*
- Achieved an excellent **99.95** percentile in **MHTCET**, competing against **0.4 million** students *(2022)*
- Selected for the **KVPY Fellowship** by DST, Government of India, with All India Rank **1627** *(2022)*

Professional Experience

Machine Learning Intern *(May 2025 - Jul 2025)*
Cognitio Analytics

AI-Powered Code Translation Research

- Conducted **market research** on SAS-to-Python translation tools, identifying capabilities and limitations
- Investigated the feasibility of adapting **LLMs** for SAS-Python code conversion via **RAG** and **fine-tuning**
- Benchmarked 20+ LLMs across critical performance metrics to identify optimal model for code translation

Automated SAS-to-Python Code Translation System

- Built an **end-to-end pipeline** to automate code translation from SAS to Python, achieving **88% accuracy**
- Conceptualized and implemented the conversion of SAS code to several intermediate representations to more effectively capture code structure and semantic information for enhanced code translation accuracy
- Deployed the pipeline as an interactive **web application** on **Gradio**, enabling seamless code conversion

Key Projects

DRONEacharya *(Apr 2023 - Aug 2023)*
Institute Technical Summer Project | ITC, IIT Bombay

- Designed and developed an aerial **vision-assisted autonomous system** combining a quadcopter to locate golf balls on a golf field and a guided rover to collect them, traversing an optimal path in minimum time
- Modified a pre-trained **TensorFlow-based model** to detect golf balls on a field and extract coordinates using a **convolutional neural network**, achieving **96% detection accuracy** on 150+ image samples
- Developed a script to create a coordinate grid of an image using the **OpenCV** computer vision library
- Implemented a **tracking system** to obtain the position of the rover from a video feed using ArUco markers

Synthetic Data Generation using GAN for Surface Defect Detection *(Mar 2024 - May 2024)*
Course Project ME228: Prof. Alankar Alankar

- **Expanded dataset size by 100%** (2,000 → 4,000 images) by developing and training a **GAN** to generate realistic synthetic surface defect samples, improving model robustness and mitigating overfitting issues
- Adapted a published model for a **Kaggle** surface defect dataset, integrating domain-specific preprocessing
- Trained a **CNN** to classify surface defects and compared its performance using the original dataset versus the augmented dataset as training data, achieving **67% accuracy** using the augmented training data

Stock Price Prediction using LSTM and RNN *(Apr 2025)*
Self Project

- Built an end-to-end **stock forecasting** pipeline using **YFinance** to fetch historical AAPL data and trained both **Recurrent Neural Network** and **LSTM** models to capture the temporal patterns in time series
- Preprocessed data with **scaling, normalization**, and sequence generation to optimize model performance
- Packaged workflow into a **modular architecture** for streamlined training, evaluation, and output storage

Super Resolution: Enhancing Images using Deep Learning — *(Dec 2023 - Jan 2024)*
Winter in Data Science | Analytics Club, IIT Bombay

- Engineered a **Generative Adversarial Network**-based super-resolution model to boost image resolution by **11.6× (120×120 → 1400×1400)**, achieving high-fidelity realistic outputs with minimal artifacts
- Acquired theoretical and practical expertise in **CNNs** and **GANs** through online courses and mentorship
- Analyzed a research paper on **Enhanced Super Resolution** and implemented the model using **PyTorch**

Credit Card Fraud Detection — *(Jul 2025)*
Self Project

- Balanced credit card fraud data using oversampling, **SMOTE**, **ADASYN** to improve detection accuracy
- Achieved **0.97 ROC AUC** using **Logistic Regression**, outperforming **Random Forest** and **XGBoost**
- Validated model robustness using comprehensive performance metrics, including **ROC, precision, recall**

Comparative Study of Image Denoising Architectures — *(Mar 2025)*
Self Project

- Implemented and trained **DnCNN** and **U-Net** based deep learning models on the CBSD68 dataset using paired noisy and clean image sets for noise removal and **image restoration** to compare their performance
- Achieved **PSNR/SSIM** scores of **73 / 0.9998** with DnCNN and **75 / 0.9999** with U-Net, reflecting a 2.7% PSNR and a 0.01% SSIM improvement, confirming the superior performance of the U-Net architecture
- Visualized denoised outputs alongside the ground truth images to qualitatively assess restoration quality

Text Classification using Transformer Encoder Model — *(Feb 2025)*
Self Project

- Implemented a **Transformer encoder** model to classify sentiment from the IMDb dataset, leveraging **TextVectorization**, embedding layers, positional encoding, **self-attention** for effective text processing
- Developed a training pipeline with tokenization, embedding layers, and encoder blocks in **TensorFlow**
- Applied **Transformer** concepts to implement attention mechanisms for effective sentiment classification

Colorful Image Colorization — *(Jul 2024)*
Self Project

- Carried out a detailed analysis of a research paper on the subject of **image colorization** using CNNs
- Utilized the L channel of the image from the **CIELab color space** as input to a machine learning model to accurately predict the a and b color channels, generating a realistic and true-to-life version of the image
- Trained a **Convolutional Neural Network** to transform a grayscale image into a full color picture

Key Courses

Mechanical Engineering	**Applied Data Science and Machine Learning**
Interdisciplinary	**A First Course in Optimization, Multi-Agent Machine Learning** †, **Computer Programming and Utilization**, Combinatorial Game Theory, Simulation Modeling and Analysis †, Biostatistics in Healthcare †
Online Courses	**Neural Networks and Deep Learning**
Miscellaneous	**Linear Algebra**, **Differential Equations**, **Calculus**

† *to be completed by Nov 2025*

Technical Proficiencies

Programming Languages	**R, SQL, Python**, C, C++, MATLAB
Software Libraries	**NumPy, Pandas, Matplotlib, scikit-learn, TensorFlow, PyTorch, Keras**, OpenCV, SciPy, Gradio, Streamlit, NetworkX, Plotly
Software Tools	**Git, GitHub, Jupyter Notebook, Weights and Biases**, Simulink, Spenvis, Ansys Mechanical, Fusion360, SolidWorks, Ubuntu, ANTLR

Extracurriculars

Competitions

- Won first prize in the National Spelling Bee in **Japan**, competing against schools across the country, and secured a spot in the finals in **Washington, US** *(2017)*
- Attained **7th rank** and a scholarship from the Maharashtra government for outstanding performance in the state scholarship exam, having scored **276/300** *(2014)*

Sports

- Completed one year of rigorous training and personal development in the **National Cadet Corps** under the guidance of personnel from the armed forces *(2022 – 2023)*
- Participated in a **5-kilometer marathon** in the NCC General championships *(2023)*